Quantitative Asset Pricing & Risk Analysis (2025 Daily Data)

This project demonstrates a quantitative financial analysis framework using daily trading data for 2025. The study applies factor-based asset pricing, time-series forecasting, and fixed-income duration modeling to examine stock return behavior and interest rate sensitivity using Python.

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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.api as sm
from statsmodels.tsa.arima.model import ARIMA

# Load dataset
data = pd.read_csv('data.csv', parse_dates=['Date'], index_col='Date')
data = data.asfreq('B')

# Compute returns
data['StockRet'] = data['Price'].pct_change()
data['ExcessRet'] = data['StockRet'] - data['RF']
data['Mkt_RF'] = data['MKT'] - data['RF']
data = data.dropna()

data.head()
```

Date	Price	MKT	SMB	HML	RF	BondYield	StockRet	ExcessRet	Mkt_RF
2025-01-02	99.372980	-0.001183	-0.032313	0.005806	0.00006	0.035183	-0.012925	-0.012985	-0.001243
2025-01-03	100.398532	0.006677	-0.010144	-0.007533	0.00006	0.035351	0.010320	0.010260	0.006617
2025-01-06	102.726875	0.015430	-0.002426	-0.017949	0.00006	0.035675	0.023191	0.023131	0.015370
2025-01-07	101.635719	-0.002142	-0.012378	-0.016175	0.00006	0.035926	-0.010622	-0.010682	-0.002202
2025-01-08	102.645191	-0.002141	0.016424	0.000581	0.00006	0.036063	0.009932	0.009872	-0.002201

Factor Regression Analysis

Fama-French Three-Factor Model regression results.

```
X = data[['Mkt_RF','SMB','HML']]
X = sm.add_constant(X)
Y = data['ExcessRet']

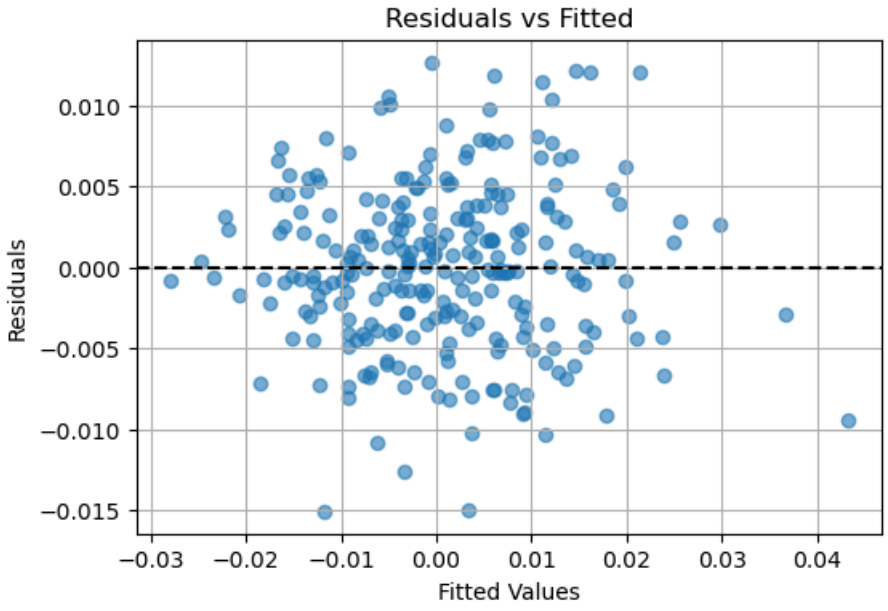
model = sm.OLS(Y, X).fit()
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              ExcessRet   R-squared:                       0.823
Model:                            OLS   Adj. R-squared:                  0.821
Method:                 Least Squares   F-statistic:                     396.0
Date:                Fri, 16 Jan 2026   Prob (F-statistic):           7.84e-96
Time:                        13:07:39   Log-Likelihood:                 1003.4
No. Observations:                 260   AIC:                            -1999.
Df Residuals:                     256   BIC:                            -1985.
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0004      0.000      1.372      0.171      -0.000       0.001
Mkt_RF         1.0885      0.033     33.151      0.000       1.024       1.153
SMB            0.3087      0.033      9.465      0.000       0.244       0.373
HML           -0.1539      0.032     -4.876      0.000      -0.216      -0.092
==============================================================================
Omnibus:                        0.070   Durbin-Watson:                   2.181
Prob(Omnibus):                  0.966   Jarque-Bera (JB):                0.013
Skew:                          -0.017   Prob(JB):                        0.993
Kurtosis:                       3.007   Cond. No.                         106.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

```python
# Residual diagnostics
fitted_vals = model.fittedvalues
resids = model.resid

plt.figure(figsize=(6,4))
plt.scatter(fitted_vals, resids, alpha=0.6)
plt.axhline(0, color='black', linestyle='--')
plt.title('Residuals vs Fitted')
plt.xlabel('Fitted Values')
plt.ylabel('Residuals')
plt.grid(True)
plt.show()
```



Time-Series Modeling and Forecasting

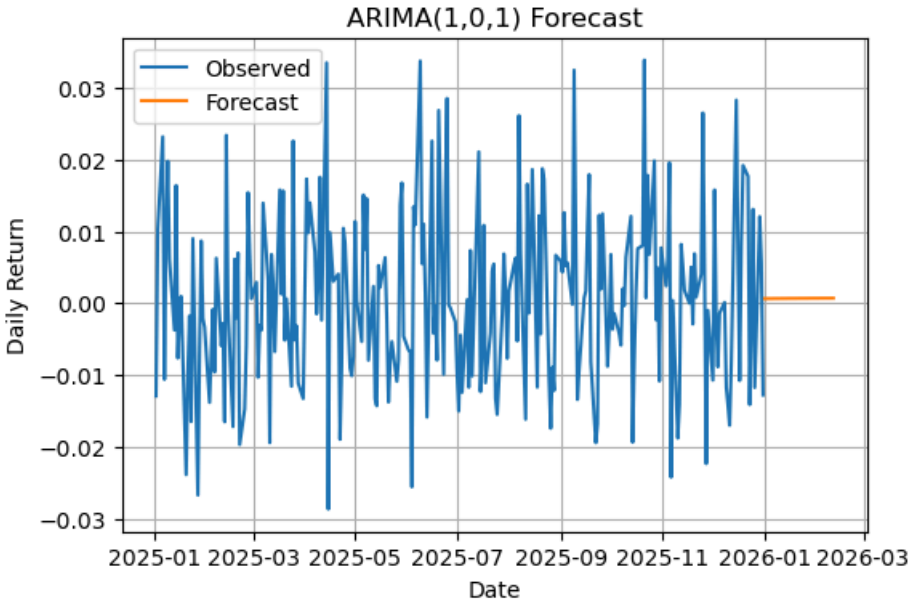
ARIMA(1,0,1) model on daily stock returns.

```python
stock_returns = data['StockRet']
arima_model = ARIMA(stock_returns, order=(1,0,1)).fit()

forecast = arima_model.get_forecast(steps=30)
forecast_index = pd.bdate_range(start=stock_returns.index[-1] + pd.Timedelta(days=1), periods=30, freq='
```

```
forecast_mean = pd.Series(forecast.predicted_mean, index=forecast_index)

plt.figure(figsize=(6,4))
plt.plot(stock_returns.index, stock_returns, label='Observed')
plt.plot(forecast_index, forecast_mean, label='Forecast')
plt.title('ARIMA(1,0,1) Forecast')
plt.xlabel('Date')
plt.ylabel('Daily Return')
plt.legend()
plt.grid(True)
plt.show()
```



Duration and Bond Price Sensitivity

Computation of Macaulay and Modified Duration and price-yield relationship.

```
face = 1000
coupon_rate = 0.05
coupon = face * coupon_rate
maturity = 10
yield0 = data['BondYield'].iloc[-1]

times = np.arange(1, maturity+1)
cashflows = np.full(maturity, coupon)
cashflows[-1] += face
dfactors = 1/((1+yield0)**times)

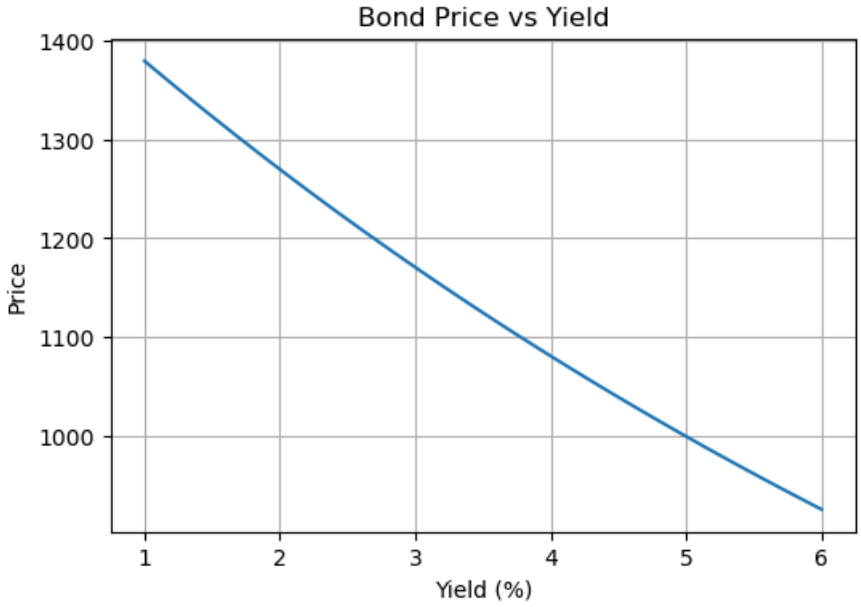
price0 = (cashflows * dfactors).sum()
weighted_times = times * (cashflows * dfactors)
macaulay_duration = weighted_times.sum() / price0
modified_duration = macaulay_duration / (1 + yield0)

print('Macaulay Duration:', round(macaulay_duration,2))
print('Modified Duration:', round(modified_duration,2))

yields = np.linspace(0.01, 0.06, 50)
prices = [ (cashflows / ((1+y)**times)).sum() for y in yields ]

plt.figure(figsize=(6,4))
plt.plot(yields*100, prices)
plt.title('Bond Price vs Yield')
plt.xlabel('Yield (%)')
plt.ylabel('Price')
plt.grid(True)
plt.show()
```

```
Macaulay Duration: 8.13
Modified Duration: 7.76
```



Conclusion

Factor Regression

The Fama-French three-factor regression effectively explained the variation in stock excess returns. The estimated factor loadings indicate the stock's sensitivity to market, size, and value risk factors, while the regression goodness-of-fit confirms that systematic risk factors account for a significant portion of return behavior.

Time-Series Modeling

The ARIMA model captured short-term dynamics in daily stock returns and provided reasonable forecasts. The fitted model demonstrates that historical return patterns contain limited but measurable predictive structure, consistent with financial market efficiency assumptions.

Duration Sensitivity

Bond duration measures were successfully computed to evaluate interest rate risk. The results confirm that bond prices move inversely with yield changes, and the duration-based sensitivity estimates provide a practical approximation of price fluctuations due to interest rate movements